SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS: **Total Marks: 20**

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.

2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.

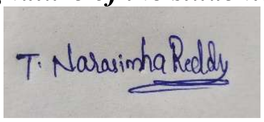
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.

4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I T. Narasimha Reddy(192425190)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

T. Narasimha Reddy - 192425190

① Difference between Machine learning and Deep learning

Introduction:

Machine learning and Deep learning are two important branches of Artificial Intelligence (AI). machine learning enables computer to learn pattern from data using algorithms, while Deep learning is a subset of Machine learning that uses multi-layered Artifical Neural Network to automatically learn Complex representation from large datasets.

parameter	Machine learning (ML)	Deep learning (DL)
Architecture	uses shallow such as Decision Trees, SVM, Random forest and Naive Bayes	use deep neural network with multi hidden layer
Feature extraction	manaual feature engineering	Automatically extract feature from raw
Training data	Faster training with lower computational	slower training
performance	suitable for structed	Best for unstructed data such as image.
Interpretability	Easier to understand and interpret	Diffcult to interpret

Architecture comparison

Input Data
↓
Data preprocessing
↓
Manual Feature Engineering
↓
Machine learning Algorithm
↓
prediction

Realworld application of Machine learning

Email spam Detection

Email services such as Gmail and outlook use machine learning algorithm to classify incoming email as spam or legitimate. The model learns from historical Email pattern and conti-nuosly improve its accuracy.

Other Application

*) Stocks Market prediction
*) weather forecasting
*) Disease prediction
*) customer prediction
*) Recommedation system.
*) stock market trading

(5)

Representation learning is a Machine learning technique in which a neural network automatically learns meaningful and useful feature ~~s~~ (representations) for raw input data. Instead of manaually designing feature, the network extract them using through multiple hidden layers.

Need for ~~re~~ Representation learning
*) Manual feature extraction is time-consuming
*) Feature quality depends on domain expertise
*) Difficult to handle complex and high-dimensional data.
*) limited performance on images, speech and text.
Representation learning overcomes these limitations by automatically extracting relevant feature, reducing.

working principle of Representation Learning
Representation learning follows a hierarchical approach where each hidden layers learns increasingly complex feature.

stage 1: Input layer
Raw data suchas images, text or audio is provided

Example:
Image pixel
Audio signal
Textwords.

Diagram of Representation learning

Raw Input Data

↓

Hidden Layer 1
(edges & lines)

↓

Hidden layer (2)

↓

Hidden layer 3
(Texture & parts)

↓

Hidden layer 4
(complete object)

↓

prediction

Advantages of Representation Learning

*) Eliminates manual feature engineering

*) Learn complex patterns automatically

*) Handle high-dimensional data efficiently

*) Improve prediction accuracy.

*) perform well on images, speech and natural language.

*) Reduce dependency on domain experts

*) Learns hierarchical feature for better generalization

③ Compare ReLu leaky ReLu and ELU

Introduction

Activation functions are one of the most important components of Artifical Neural Networks (ANNs) and Deep learning models. They introduce non-linearity into the network, enabling it to learn complex patterns and relationships from data. with out any activation function, a neural network, enabling it to learn complex patterns linear model regardless of the number.

Needs for Activation functions

*) A neural network behaves like a linear regression model
*) multiple hidden layer cannot learn complex nonlinear.
*) The network cannot classify images, speech, or text efficiently
*) Learning capability becomes very limited.
*) learning capability becomes very limited.

Mathematical formula

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

working principle

1) If the input value is positive, the output is equal to the input

2) If the input value is negative, the output becomes zero

Input	output
-8	0
-3	0
-1	0
0	0
2	2
5	5
10	10

Leaky Relu (LRelu)

Leaky Relu is an improved version of Relu that allows a small non-zero output for negative input values instead for making them zero

mathematical formula

$$f(x) = \begin{cases} \alpha x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

α (alpha) $= 0.01$

④ concept of unsupervised learning in neural network

Introduction:

Artifical intelligence (AI) enables machines to perform task such as learning, reasoning, prediction, and decision making. one of the important branches.

Among these unsupervised learning is widely used when labeled data is unavailable. Instead of learning from predenfined target outputs, the model analyzes unlabeled data.

unsupervised learning in Neural Networks

unsupervised learning is a type of machine learning in which the neural network is trained using unlabeled data.

characteristic of unsupervised learning

*) uses unlabeled dataset
*) Automatically discovers hidden structure
*) performs feature extraction
*) Identifies similarities and difference
*) Helps in anomaly detection
*) useful when labeled data is unavailable.

working process of unsupervised learning

Raw unlabeled data

↓

Data preprocessing

↓

Neural Network

↓

pattern Discovery

↓

feature learning

↓

useful Representation

Advantages of unsupervised learning

*) Does not require labeled data

*) Reduce manual effort

*) Discover hidden relationships.

*) useful for large dataset.

*) perform automatic feature learning

Disadvantages of unsupervised learning

*) Diffcult to evaluate results

*) lower interpretability

*) sensitive to noisy data

*) Require careful parameter tuning